

AUDITABLE DENOISING CONTROL VIA OBSERVATION-RELATIVE TOTAL VARIATION BUDGETS

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ABSTRACT

Denoising strength alone does not specify how much observed variation an output may remove. This paper presents an observation-relative total variation (TV) budget for auditable denoising control. A retraining-free controller seeks maximal feasible proposal retention along the observation–proposal segment. Compliant proposals pass unchanged; otherwise, convexity establishes a feasible prefix, enabling bisection despite nonmonotone segment TV. Experiments on 16 Kodak images and 21 PolyU filename groups show mean PSNR gains over validation-selected fixed blending of 0.79 and 0.53 dB at the reported budgets. On synthetic weak-line detection, TV control improves mean AP over the proposal by 0.0431, mainly under denoising mismatch, and nearly matches validation-selected fixed blending in AP with lower background RMSE. Independent float64 audits confirm compliance within 10^{-12} for all 1,380 evaluated restoration controller outputs. Twelve-step bisection differs from 28 steps by less than 0.0045 dB in PSNR while roughly halving control time at two reported operating points.

Index Terms— image denoising, total variation, output control, auditability, restoration

1. INTRODUCTION

Image denoising estimates a clean image from a noisy observation and is fundamental to imaging and inverse problems [1]. A denoising front-end may also need an explicit allowance on how much observed variation it removes. A model strength setting alone cannot enforce this requirement: the same setting can remove different amounts of variation from different images. This creates a gap between controlling denoiser behavior and constraining its output.

Total variation (TV) aggregates spatial gradient magnitudes and has long been used in restoration through regularization [2], constrained optimization [3, 4], and TV-ball projection [5]. Our formulation instead places a lower bound on output TV relative to the observation, specifying how much aggregate variation the output may remove. Normalizing by the observation’s maximum removable TV gives the allowance a dimensionless fractional interpretation.

Existing controllable restoration methods interpolate network weights [6], modulate features [7], or weight learned residual connections [8]. Real-time controllable denoising (RCD) [9] enables noise-map interpolation using a trained output head. Instruction-based restoration [10] and PRISM [11] provide text-based control, while UnSCAR [12] adjusts expert weights. Diffusion methods expose further reconstruction tradeoffs [13]. These controls adjust restoration behavior rather than enforce a prescribed observation-relative TV-removal budget. Plug-and-play methods incorporate denoisers into iterative reconstruction [14, 15, 16, 17], whereas we control an already completed output.

We therefore ask: *given a noisy observation and a completed denoising proposal, what is the largest proposal fraction that satisfies an observation-relative TV-removal budget?* The challenge is that TV need not vary monotonically along the observation–proposal segment. We show that checking the proposal first resolves the search geometry: a feasible proposal is optimal, while an infeasible proposal leaves a feasible prefix interval. This supports bisection toward maximal feasible retention, with numerical acceptance checked on the returned array. Figure 1 summarizes the workflow, from budget specification to search and independent audit.

Our contributions are:

- An observation-relative TV budget that converts a dimensionless removal allowance into an explicit output-TV floor.
- A segment-feasibility characterization enabling retraining-free search for maximal feasible proposal retention, even with nonmonotone segment TV.
- An evaluation isolating gains from per-image allocation and boundary localization, demonstrating weak-structure detectability recovery under denoising mismatch, with independent output audits and a precision–cost analysis.

2. PROPOSED METHOD

2.1. Observation-Relative TV Budget

For an observation f and output u in $[0, 1]^{H \times W \times C}$, we use vector-valued TV [18] and define its normalized smoothed form as

$$V_\epsilon(u) = \frac{1}{HW} \sum_{i,j} \sqrt{\|\nabla u_{i,j}\|_2^2 + \epsilon^2}, \quad \epsilon = 10^{-8}. \quad (1)$$

We compute forward differences with zero outward boundary differences, and the norm combines both spatial directions and all channels. With this convention, $V_\epsilon(u) \geq \epsilon$, with equality exactly when each channel is spatially constant. This lower bound gives us a natural reference for the maximum amount of TV that can be removed from f .

Definition 1 (Observation-Relative TV Budget). For fixed f , the net removed variation is

$$R(u; f) = \max\{0, V_\epsilon(f) - V_\epsilon(u)\}. \quad (2)$$

Given $\beta \in [0, 1)$, an output is feasible if

$$R(u; f) \leq B_\beta(f), \quad B_\beta(f) = \beta[V_\epsilon(f) - \epsilon]. \quad (3)$$

The parameter β therefore has a direct interpretation: it specifies the permitted fraction of the maximum removable TV, $V_\epsilon(f) - \epsilon$. We

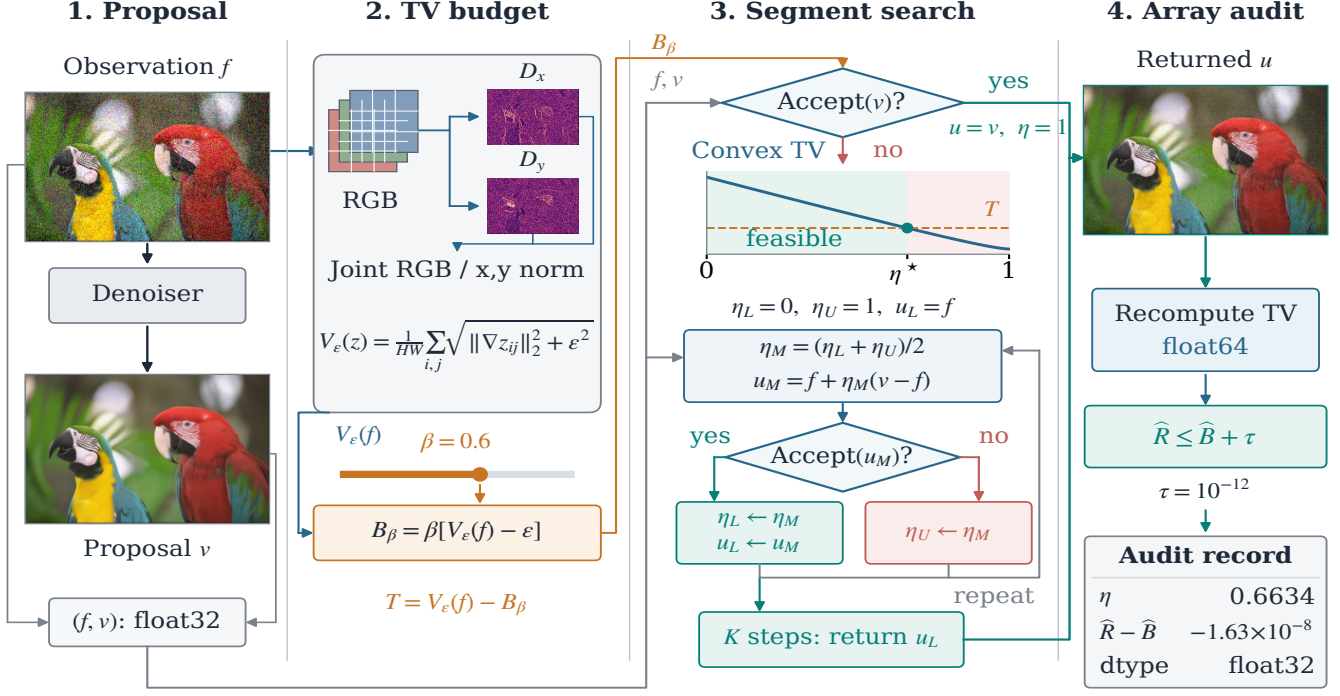


Fig. 1. Overview of the TV-budget controller.

can also express the same constraint as a lower bound on the output TV:

$$V_\epsilon(u) \geq (1 - \beta)V_\epsilon(f) + \beta\epsilon. \quad (4)$$

For nonconstant f , this exact floor excludes constant outputs whenever $\beta < 1$. At $\beta = 0$, no net TV reduction is allowed, while extending the definition to $\beta = 1$ admits every output. Constant observations impose no restriction. Importantly, this budget limits aggregate net removal; it does not guarantee local detail preservation and does not prevent TV increases.

2.2. Maximal Feasible Proposal Retention

With the budget in place, we next ask how much of a given proposal we can retain. For a clipped proposal $v \in [0, 1]^{H \times W \times C}$, we blend along the segment from f to v and seek the largest feasible proposal weight:

$$\begin{aligned} u_\eta &= f + \eta(v - f), \\ \eta^* &= \max\{\eta \in [0, 1] : R(u_\eta; f) \leq B_\beta(f)\}. \end{aligned} \quad (5)$$

A maximum exists because the feasible set is closed and contains $\eta = 0$.

Proposition 1 (Segment Feasibility). *If v is feasible, then $\eta^* = 1$. Otherwise, the feasible coefficients form $[0, \eta^*]$, even when $V_\epsilon(u_\eta)$ is nonmonotone.*

Proof. A feasible v attains the largest coefficient, $\eta = 1$. Otherwise, let $T = V_\epsilon(f) - B_\beta(f)$ and $g(\eta) = V_\epsilon(f + \eta(v - f))$. The function g is convex and continuous, with $g(0) \geq T$ and $g(1) < T$. Its strict

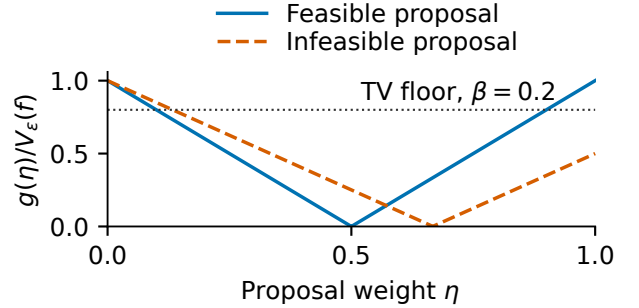


Fig. 2. Nonmonotone segment TV for $f = (0.1, 0.9)$ repeated in RGB. Proposal $(0.9, 0.1)$ has a feasible endpoint but an infeasible middle; $(0.7, 0.3)$ has a feasible prefix supporting bisection.

sublevel set $\{\eta \in [0, 1] : g(\eta) < T\}$ is an interval containing 1 but not 0. Continuity makes this interval $(\eta^*, 1]$, leaving the feasible prefix $[0, \eta^*]$. $\square$

This result also explains why we check the proposal before starting the search: a feasible endpoint can coexist with infeasible intermediate blends (Fig. 2). Maximal retention has a simple geometric interpretation as well. It minimizes distance to v among feasible segment points, since $\|u_\eta - v\|_F = (1 - \eta)\|v - f\|_F$. For fixed endpoints, increasing β enlarges the feasible set, so η^* is nondecreasing.

Algorithm 1 Budget-Constrained Proposal Retention

Require: Prepared f, v , budget β , iterations K **Ensure:** Accepted (u, η)

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1: Compute  $\hat{B}_\beta(f)$ 
2: if Accept( $v$ ) then
3:   return ( $v, 1$ )
4: end if
5:  $\eta_L \leftarrow 0, \eta_U \leftarrow 1, u_L \leftarrow f$ 
6: for  $k = 1, \dots, K$  do
7:    $\eta_M \leftarrow (\eta_L + \eta_U)/2$ 
8:    $u_M \leftarrow \text{float32}(f + \eta_M(v - f))$ 
9:   if Accept( $u_M$ ) then
10:     $\eta_L \leftarrow \eta_M, u_L \leftarrow u_M$ 
11:   else
12:     $\eta_U \leftarrow \eta_M$ 
13:   end if
14: end for
15: return ( $u_L, \eta_L$ )
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2.3. Search and Numerical Acceptance

We turn this characterization into the simple procedure in Algorithm 1. We first check the proposal and return it unchanged if accepted. Otherwise, we bisect the segment, retaining the last accepted array. In floating-point arithmetic, we use the acceptance rule

$$\text{Accept}(u) \iff \hat{R}(u; f) \leq \hat{B}_\beta(f) + \tau, \quad (6)$$

where hats denote floating-point evaluation and $\tau = 10^{-12}$.

With exact arithmetic and feasibility decisions ($\tau = 0$), bisection preserves a bracket containing η^* , yielding

$$0 \leq \eta^* - \eta_L \leq 2^{-K}. \quad (7)$$

For a rejected proposal, the search requires K midpoint TV evaluations, each linear in the number of image values.

In implementation, we distinguish this exact-arithmetic bound from acceptance of the actual returned array. We clip f, v to $[0, 1]$, store endpoints and blends in float32, and compute TV in float64. We check acceptance on the retained array before file quantization, rather than on a reconstruction from its weight. Thus, (7) describes exact-arithmetic accuracy, while (6) checks numerical compliance of the returned array. In the experiments, we independently audit these arrays in float64.

3. EXPERIMENTS

3.1. Experimental Setup

Data and proposals. Kodak [19] images 01–08 serve as validation and 09–24 as test images, resized to a maximum side of 256 with Gaussian noise $\sigma = 25/255$. PolyU [20] uses one crop per filename group: eight validation and 21 test groups; scene-level independence is unverified. All controllers share observations and clipped color DRUNet [21] proposals. For PolyU, validation PSNR selects the noise-map level 15/255 from $\{1, 2, 3, 5, 7, 10, 15, 25\}/255$.

Controllers and evaluation. We compare bisection, fixed blending, per-image halving, and coarse-grid search at tolerance 10^{-12} . Fixed blending maximizes validation PSNR over $\{0, 0.005, \dots, 1\}$ subject to every validation budget, then uses the shared coefficient without test-time fallback. Halving and grid search test 1, 1/2, $\dots$ and 1, 0.95, $\dots, 0$ until acceptance. Bisection uses 28 steps, with

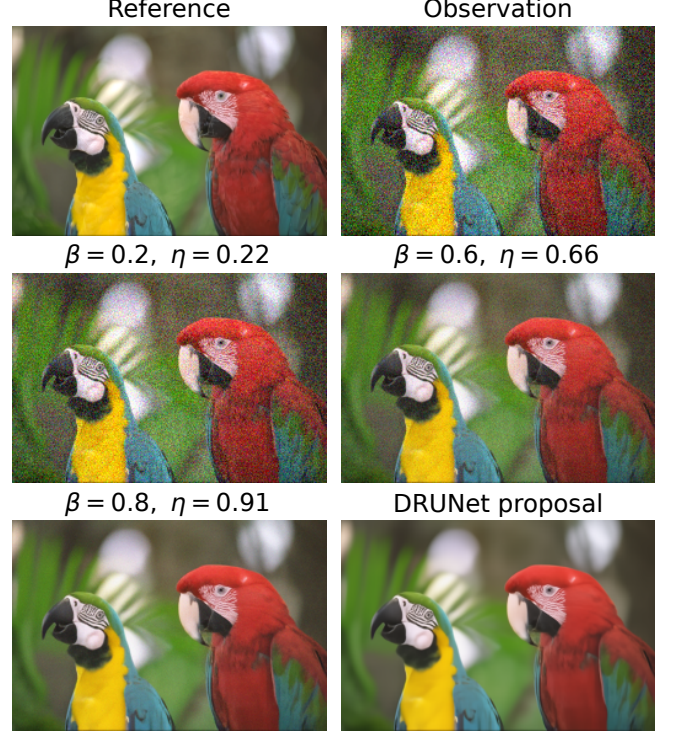


Fig. 3. Budget-controlled outputs for Kodak kodim23 at $\sigma = 25/255$, using the same observation and DRUNet proposal. Full resized images use identical display scaling.

8 and 12 in ablations. Budgets are $\beta \in \{0.2, 0.4, 0.6, 0.8\}$, plus a separate $\beta = 0.05$ setting. We report mean per-image RGB PSNR and SSIM [22, 23], with data range 1 and no boundary cropping. Paired bootstrap 95% percentile intervals use 10,000 image resamples, resampling selected crops for PolyU. Restoration comparisons are exploratory on previously inspected test sets.

3.2. Per-Image Allocation and Robustness

We first examine what per-image control adds when the denoising proposal is held fixed (Table 1). Kodak input/proposal PSNR is 20.42/31.79 dB. Adaptive blending improves PSNR at all four budgets, with a 0.79 dB gain at $\beta = 0.6$. Figure 3 makes this control concrete: increasing the TV-removal budget retains more of the same DRUNet proposal.

On PolyU, input/proposal PSNR is 35.65/37.96 dB, and adaptive gains reach 0.53/0.32 dB at $\beta = 0.2/0.4$. As the budget relaxes, the controller naturally leaves more proposals unchanged: 18 of 21 at $\beta = 0.6$ and all 21 at $\beta = 0.8$. Fixed blending is then up to 0.07 dB better, reflecting the distinction between maximal feasible retention and optimal reference PSNR. Both methods satisfy every test budget in Table 1.

These gains persist across noise levels. On Kodak, we vary $\sigma \in \{15, 25, 50\}/255$ with matching DRUNet maps, unchanged splits, and per-image seeds. Adaptive gains remain positive at both reported budgets (Table 2). At $\sigma = 50/255$, fixed blending violates one of 16 test budgets at each of $\beta = 0.4$ and 0.8, whereas all adaptive outputs remain feasible.

Table 1. DRUNet-based test results. $\tilde{\eta}$: median adaptive retention; PSNR/SSIM: per-image means; Δ PSNR: adaptive minus fixed (dB), with paired 95% CIs. No test budget violations occur.

Dataset	β	Adaptive blend			Fixed blend		Δ PSNR [95% CI]
		$\tilde{\eta}$	PSNR	SSIM	η	PSNR / SSIM	
Kodak	0.2	0.225	22.44	0.446	0.210	22.28 / 0.439	0.17 [0.12, 0.21]
	0.4	0.453	25.02	0.553	0.420	24.56 / 0.533	0.46 [0.35, 0.58]
	0.6	0.689	28.35	0.709	0.640	27.56 / 0.672	0.79 [0.60, 0.99]
	0.8	0.953	31.52	0.877	0.865	30.90 / 0.844	0.62 [0.50, 0.74]
PolyU	0.2	0.419	37.13	0.949	0.235	36.60 / 0.938	0.53 [0.43, 0.64]
	0.4	0.933	37.75	0.962	0.480	37.43 / 0.955	0.32 [0.16, 0.46]
	0.6	1.000	37.93	0.964	0.750	37.98 / 0.965	-0.06 [-0.17, 0.06]
	0.8	1.000	37.96	0.965	0.910	38.04 / 0.966	-0.07 [-0.12, -0.03]

Table 2. Kodak noise sensitivity. Δ : adaptive-minus-fixed PSNR (dB). Last column: maximum fixed violation rate across four main budgets.

255σ	$\beta = 0.2$		$\beta = 0.6$		Max. fixed viol. (%)
	$\tilde{\eta}$	Δ	$\tilde{\eta}$	Δ	
15	0.256	0.35	0.805	1.21	0.00
25	0.225	0.17	0.689	0.79	0.00
50	0.206	0.07	0.622	0.30	6.25

Table 3. Kodak proposal sensitivity at $\sigma = 25/255$. Adaptive violations are zero. DRUNet results are in Table 1.

Proposal	β	$\tilde{\eta}$	PSNR	Δ PSNR [95% CI]
Wavelet	0.2	0.227	22.18	0.15 [0.11, 0.19]
Wavelet	0.6	0.707	25.91	0.37 [0.29, 0.45]
NLM	0.2	0.234	22.33	0.17 [0.14, 0.20]
NLM	0.6	0.707	26.89	0.47 [0.39, 0.55]

The same pattern extends beyond DRUNet. At $\sigma = 25/255$, adaptive blending improves both BayesShrink [24] and non-local means (NLM) [25] proposals (Table 3). BayesShrink uses channel-wise db1 wavelets with four levels and soft thresholding. NLM uses 5×5 patches, search distance 6, $h = 0.85\sigma$, fast mode, and its noise standard deviation parameter set to zero, disabling noise-variance correction. Fixed wavelet blending violates one of 16 budgets at each reported setting; all other outputs remain feasible.

3.3. Weak-Structure Detectability under Denoising Mismatch

Beyond restoration fidelity, we ask whether budget control can recover weak structures suppressed by denoising. We generate 40 validation and 100 test geometries containing bright, antialiased lines of width 1–3 pixels and contrast 0.03–0.12 in 256×256 images. Grayscale images are replicated to RGB and corrupted by independent channel-wise Gaussian noise at $\sigma \in \{15, 25\}/255$, with three repeats per level. DRUNet receives noise maps $a\sigma$, where $a \in \{1, 1.5, 2\}$. A fixed Sato detector at scales (1, 2, 3) is scored by per-image average precision (AP) against geometric masks. Here, validation macro AP across all conditions selects one shared coefficient $\eta = 0.76$ for unconstrained fixed blending and one shared budget $\beta = 0.8$ for TV control, both frozen for testing.

Figure 4(a) shows mismatch-dependent recovery: AP changes

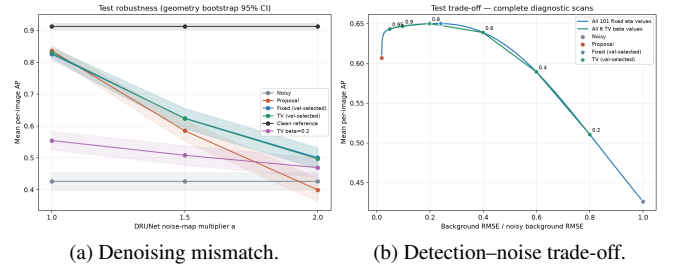


Fig. 4. Weak-structure detection on 100 held-out test geometries. (a) Mean AP versus noise-map multiplier a ; shading indicates geometry-bootstrap 95% CIs. (b) Complete fixed-blend and TV-budget scans, averaged over noise levels, repeats, and mismatch conditions. Stars mark validation-selected shared operating points; $\beta = 0.2$ is also labeled.

relative to the proposal are -0.0063 , $+0.0383$, and $+0.0974$ for $a = 1, 1.5$, and 2 , respectively. Overall, TV control raises test macro AP from 0.6070 to 0.6501, a gain of 0.0431 (95% CI [0.0397, 0.0467]; 10,000 bootstrap resamples of clean geometries retaining all paired conditions). At validation-selected operating points (Fig. 4(b)), TV control nearly matches fixed blending in AP (0.6501 versus 0.6502) with a lower background RMSE ratio (0.1975 versus 0.2410, relative to the noisy observation).

3.4. Boundary Accuracy and Control Cost

Having examined what budget control achieves, we turn to the precision and cost of implementing it. Against per-image grid search, 28-step bisection gains 0.388 dB (95% CI [0.286, 0.490]) on Kodak at $\beta = 0.6$ and 0.091 dB ([0.053, 0.134]) on PolyU at $\beta = 0.2$, recovering proposal content left unused by the coarse grid. On PolyU, the gain narrows to 0.028 dB at $\beta = 0.4$ (CI includes zero); at $\beta = 0.8$, all per-image methods return the proposal.

Timing uses Python 3.14.4/NumPy 2.5.3 with four OpenBLAS and OpenMP threads. After one warm-up per controller and dataset, each image–budget–controller condition is timed three times in randomized order; condition medians are averaged across images. Timing includes budget construction, search, and a separate float64 audit, without cross-call TV caching, but excludes proposal generation and scoring.

Full 28-step precision is not needed to preserve these gains (Fig. 5). Across 276 conditions, maximum absolute PSNR differences

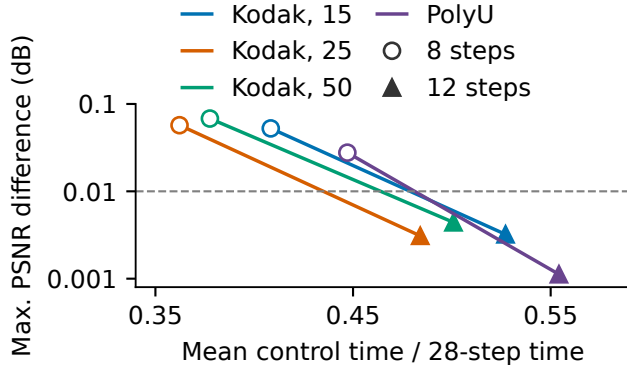


Fig. 5. Precision–cost trade-off relative to 28-step bisection across four budgets. Time is the ratio of mean condition-median times; quality is the maximum per-image absolute PSNR difference across the same conditions. Kodak labels denote 255σ .

from 28 steps stay below 0.0045 dB for 12 steps and 0.0681 dB for eight. At Kodak $\beta = 0.6$ and PolyU $\beta = 0.2$, 12 steps roughly halve control time: 35.87 to 17.19 ms and 253.51 to 124.79 ms. Grid search takes 10.36/118.53 ms but loses 0.386/0.090 dB relative to 12 steps.

Crucially, this reduction in search effort preserves budget compliance: all 1,380 outputs from halving, grid search, and 8-, 12-, and 28-step bisection pass the independent 10^{-12} audit across 276 DRUNet conditions ($16 \times 3 \times 4$ Kodak; 21×4 PolyU).

4. CONCLUSION

Observation-relative TV budgets provide an explicit, auditable constraint on net variation removal. Segment feasibility enables a simple, retraining-free search for maximal feasible proposal retention. Experiments isolate gains from per-image allocation and boundary localization, with independently audited compliance of the returned arrays. At the reported operating points, 12 steps nearly match 28-step PSNR while roughly halving control time.

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